

QUEUE

Q1) Monk and Multiplication (Product of 3 Largest Integers)

DS	Session	Queue	Question Information	Level 1	Challenge 51
Problem Description The Monk recently learnt about priority queues and requested his teacher for a fun challenge to solve. As a result, his teacher devised a simple task. He now possesses A, which is an integer array. He wishes to discover the product of the greatest, second largest, and third largest integers in the range $[1, i]$ for each index i. Note: Two numbers can be the same value-wise but they should be distinct index-wise. Constraints: $1 \leq N \leq 100000$ $0 \leq A[i] \leq 1000000$ Input: The first line contains an integer N, denoting the number of elements in the array A. The next line contains N space separated integers, each denoting the ith integer of the array A. Output: Print the answer for each index in each line. If there is no second largest or third largest number in the array A upto that index, then print "-1", without the quotes.					

CODE:

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#include <stdio.h>

typedef struct { long long d[4]; int f, r, sz; } Queue;

void push(Queue *q, long long x) { q->d[q->r = (q->r + 1) % 4] = x; q->sz++; }
long long pop(Queue *q) { q->sz--; long long v = q->d[q->f]; q->f = (q->f + 1) % 4; return v; }

int main(void) {
    int n;
    if (scanf(format: "%d", &n) != 1) return 0;
    Queue q = {.f = 0, .r = -1, .sz = 0};

    for (int i = 0; i < n; i++) {
        long long x, min = 2e18;
        scanf(format: "%lld", &x);
        if (q.sz < 3) push(q, x);
        else {
            for (int k = 0; k < 3; k++) {
                long long v = pop(q);
                if (v < min) min = v;
                push(q, x);
            }
            int rep = 0;
            for (int k = 0; k < 3; k++) {
                long long v = pop(q);
                push(q, x);
            }
            if (i < 2) printf(format: "-1\n");
            else {
                long long p = 1;
                for (int k = 0; k < 3; k++) { long long v = pop(q); p *= v; push(q, x); }
                printf(format: "%lld\n", p);
            }
        }
    }
    return 0;
}

```

OUTPUT:

```

[suraj@suraj-83gs test]$ ./main
5
1 2 3 4 5
-1
-1
6
24
60

```